Track 3: The Microfilm (Converging Lenses)

Gradient Index Optics using Ray Marching
ENSTA – DFI-MI2
Department of Basic Computer Science Training
Academic Year 2025–2026

Definition of a Lens:

A lens is a transparent optical element, usually made of glass or plastic, that refract sligh t and changes its direction to form real or virtual images.

› Types of Lenses:

Lenses are divided into two main types:

- **1. Converging (Convex) Lens — brings light rays together at a focal point.**
- **2. Diverging (Concave) Lens — spreads light rays apart.**

(in our case we going to study just convergent lens)

› Converging Lens:

A converging lens is thick at the center and thin at the edges. It can form real or virtual images depending on the object's position relative to the lens. One of its most well-known applications is the magnifying lens

› Magnifying Lens :

The magnifying lens is a type of converging lens used to enlarge small objects,making fine details visible.

The magnification can be adjusted by changing the distance between the object and the lens. And we have 2 cases, generate a magnified real or virtuelle image .

Magnifying lenses are used in microscopes, projection .

(In our case we going to focus on projection)

PART OF QUESTION :

Q1 Focal Length and Optical Power:

First difintion of focal length :

Focal Length (f): Distance between the lens center and its focal point (where parallel light rays converge)

Measured in meters (m) or centimeters (cm).

Short focal length → lens bends light strongly; long focal length → lens bends light weakly. Indicates where the image will focus.

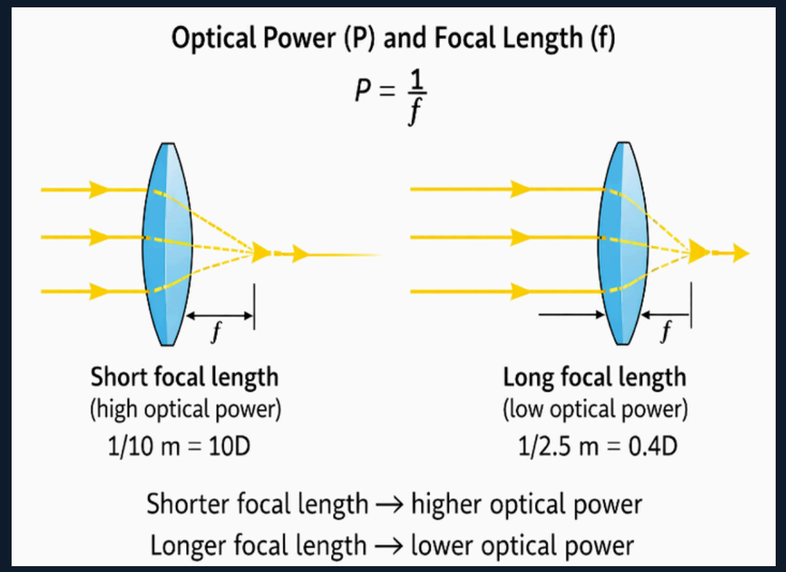
Optical Power (P):

Second difintion of optical power :

Measures how strongly a lens bends light.

Measured in diopters (D); 1 D = 1/m.

Higher optical power → stronger bending of light.



Third Mathematical Relationship:

P=1/f =>f=1/p so in our exercice f=1/10=0.01 m

Optical power and focal length are inversely proportional.

Larger optical power → shorter focal length;

smaller optical power → longer focal length.

Note:

For converging (convex) lenses, both focal length and optical power are positive.

Q2 :Claculting the exact object postion

First, OA represents the algebraic distance between the object and the lens.If OA is negative, the object is located on the left side of the lens (the object side).If OA is positive, the object is located on the right side of the lens.

By comparing the value of $|OA|$ with the focal length f, we can determine the nature of the image (real or virtual) and whether it will be magnified, reduced, or remain the same size.

Using thin lens equation :

$$1/OA' - 1/OA = 1/f'$$

Since the lens has an optical power P = +10 D, its focal length is:

$$f' = 1/P = 0.10 \text{ m}$$

The screen (wall) is at OA' = +1.50 m.

Substituting into the lens equation and solving gives:

$$OA \approx -0.107 \text{ m}$$

Theoretical explanation (short):

The focal length is positive because the lens is converging.

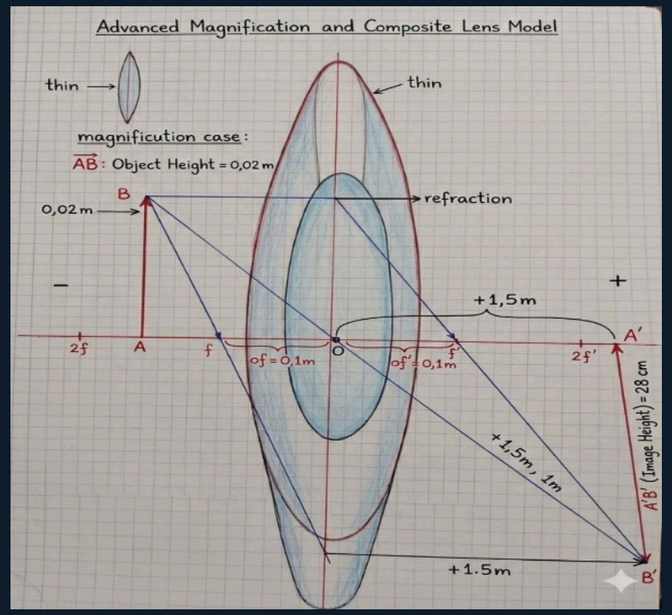
The image distance OA' is positive because the image is real and formed on the screen.

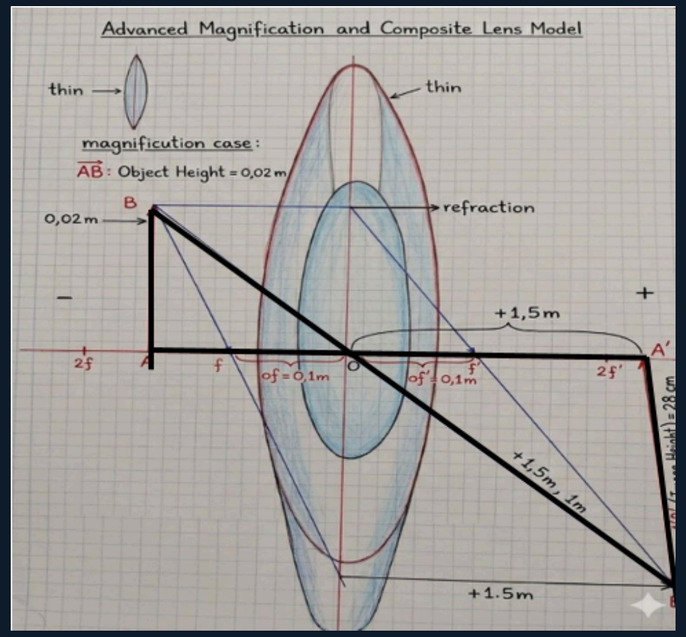
The object distance OA is negative, which means the object is placed on the object side of the lens (in front of it).

Q3 : The magnification

Transverse magnification measures how much larger or smaller the image is compared to the object in the direction perpendicular to the principal axis, and also indicates whether the image is upright or inverted

M = height of image /height of object = AB/A'B' =OA/O'A' this last using thales theorem





Where:

M = magnification (tells you the size and orientation of the image)

OA' = distance from the lens to the image

OA = distance from the lens to the object

AB=height of the object

A'B'=height of the image

What M tells us (making difference between magnification and minimaizing):

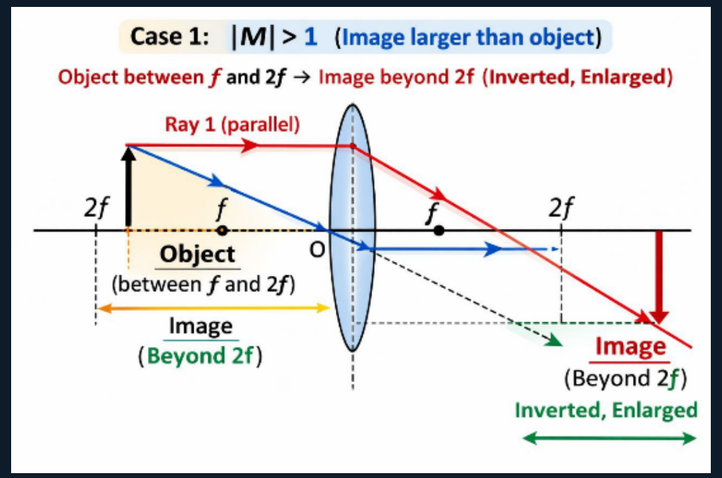
Remark : in projection the image always will construct by the intersection of many refracted light rays from the object,this image will be always inverted in basic optics

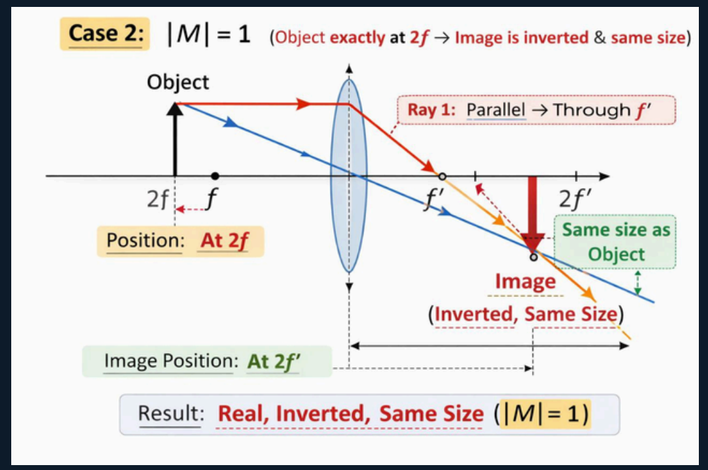
THE THREE CASES ARE:

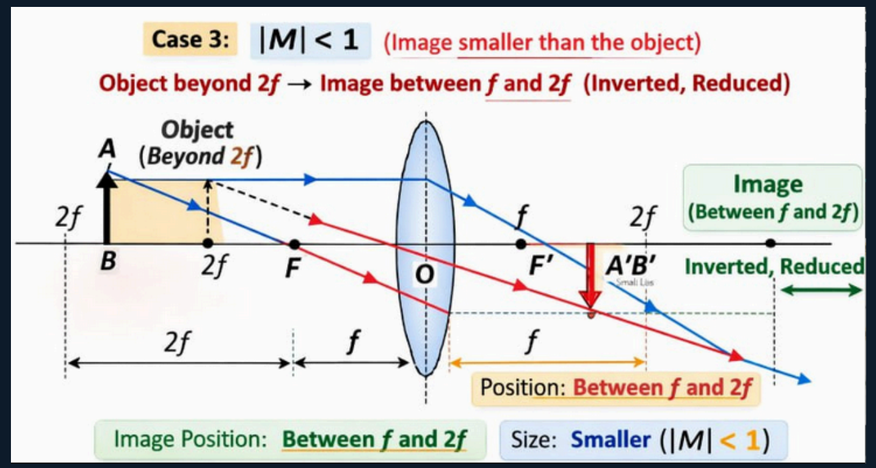
M < 0 → image is inverted (upside down)

M > 0 → image is upright (same way up) this case is in virtual image

|M| > 1 → image is larger than the object(this case happen when the object is between f and 2f)







|M| < 1 → image is smaller than the object (this case happen when the object is far from 2f)

|M| = 1 → image is the same size as the object(this case happen if the object is exactly in postion of 2f)

In the exercice : M =OA'/OA=1.5/-0.107=-14 So image is inverted larger then the object

Question 4: Height and Orientation of the Projected Image

The transverse magnification is defined as:

$M = A'B' / AB$.

We already know:

* $AB = 2.0$ cm
* $M = -14$

The formula to calculate the image height is:

$A'B' = M \times AB$

Substituting the values:

$A'B' = (-14) \times (2.0 \text{ cm})$

$A'B' = -28$ cm

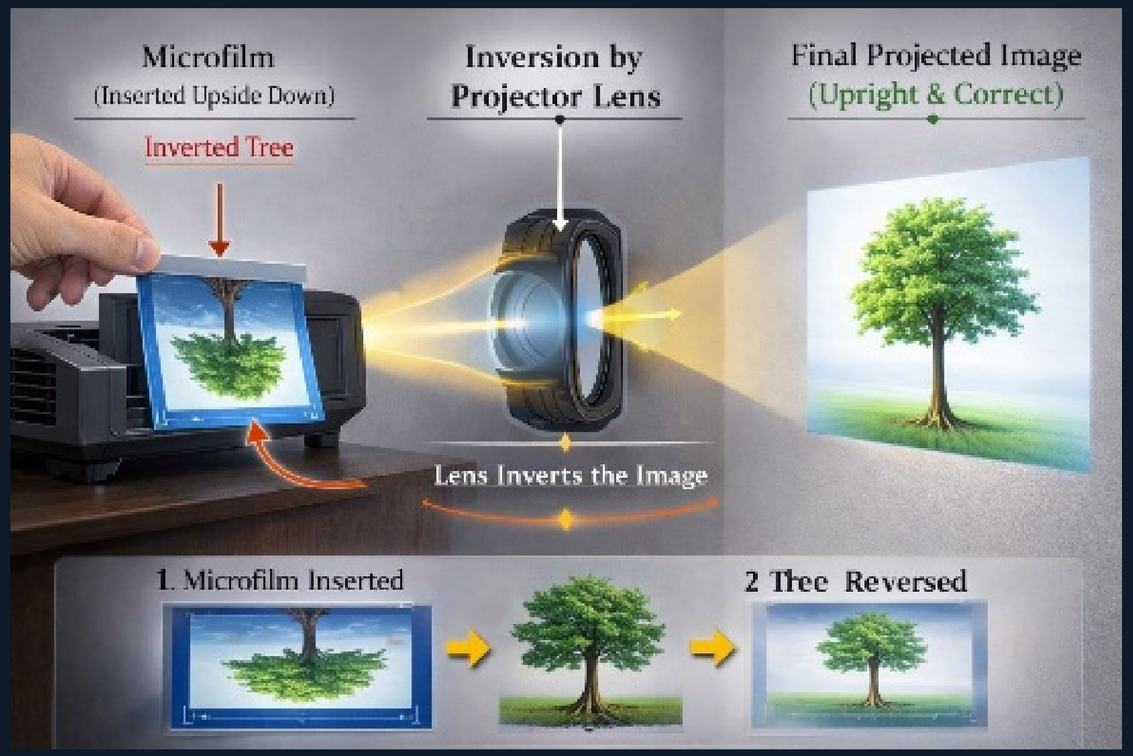
The absolute value gives the physical height of the image:

$|A'B'| = 28$ cm

Therefore, the projected image is 28 cm tall.

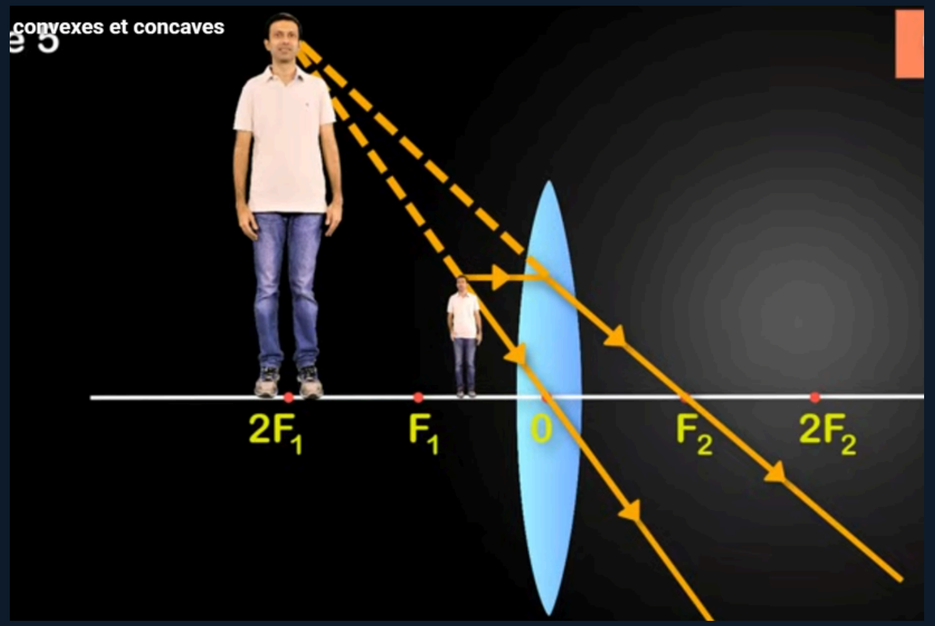
2. Orientation of the Microfilm :

Because the lens inverts the image:
If the microfilm is inserted right-side up, the projected image will appear upside down on the wall.
To obtain a correctly oriented and readable blueprint on the wall, the microfilm must be inserted upside down.
The lens will invert it during projection, and the final image will appear upright



Additional point about virtual image:

virtual image with converge lens is formed when light rays do not actually intersect but instead diverge after reflection or refraction, appearing as if they originate from a point behind the mirror or lens; therefore, it cannot be projected onto a screen and is usually upright and magnified in converge lens . As application of that “the magnifying glass”



Conclusion :

the magnifying glass is considered a fundamental optical tool that operates based on the principle of light refraction to produce an enlarged image of nearby objects, allowing the observation of fine details that cannot be seen with the naked eye. This tool has contributed significantly to scientific research, education, and various practical applications. Therefore, understanding how a magnifying glassworks highlights the importance of physical principles in developing tools that enhance human observation and precision. Thank you for your attention.

Thank You